



Chris Kraft passes away at 95

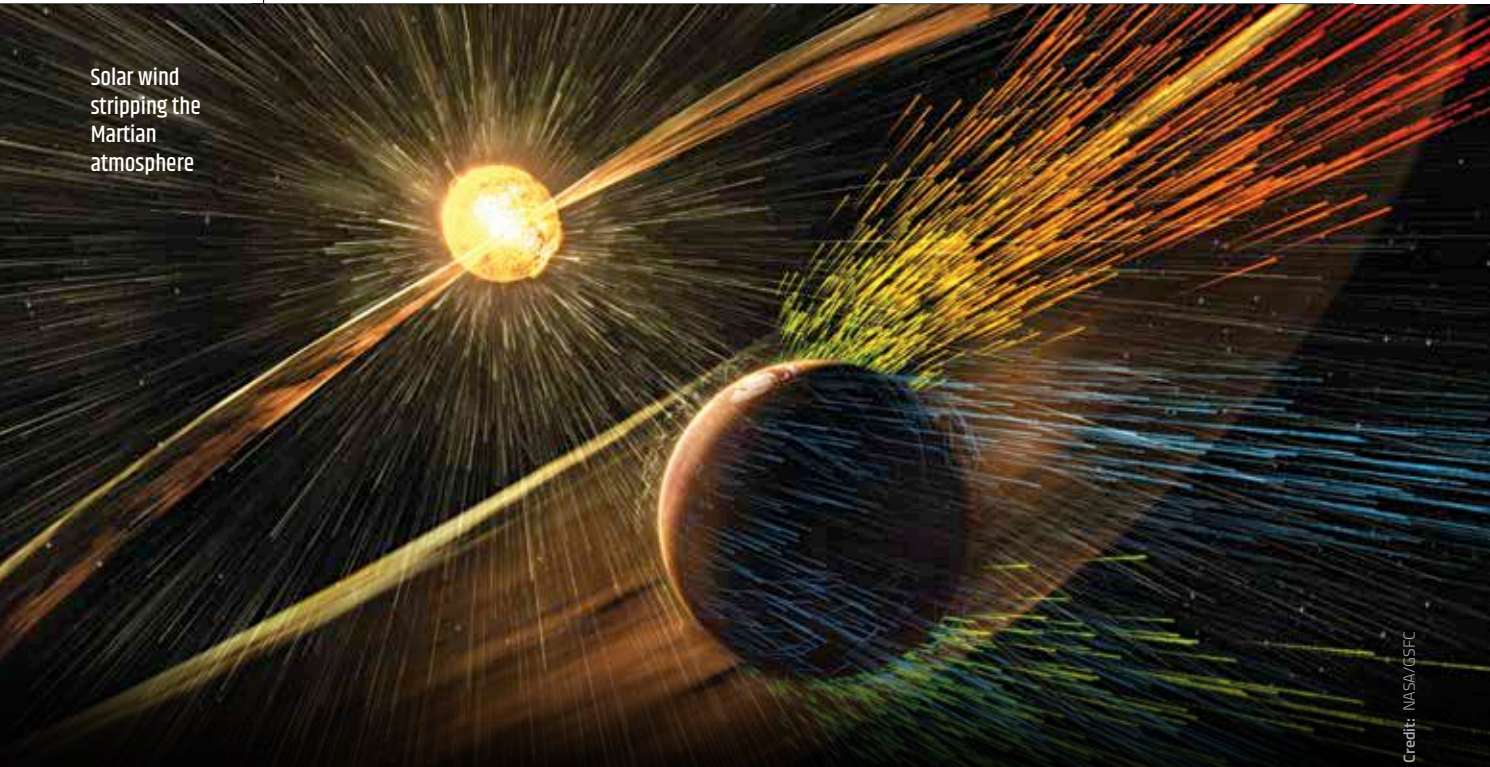
NASA's first flight director, Chris Kraft, passed away at 95, just days before the anniversary of the first Moon landing. <https://digit.in/aug19-15>



More water on the moon!

India's follow-up lunar mission, Chandrayaan 2, has found more ice on the Moon that previously speculated. <https://digit.in/aug19-16>

Solar wind stripping the Martian atmosphere

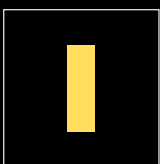


Credit: NASA/OSFC

TERRAFORMING MARS

Apart from the Earth, Mars is the most habitable planet in the Solar System, and making the entire planet habitable has been one of the most popular themes of Science Fiction.

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In Kim Stanley Robinson's Mars Trilogy, the red planet is painstakingly terraformed

over a duration of two centuries. The obstacles that have to be overcome are not only environmental, but also political. A number of games, including Surviving Mars, allow you to experience the process of terra-

forming Mars. The idea is to make Mars more similar to the Earth by providing it with a thick atmosphere, and sufficient liquid water on the surface to initiate a water cycle.

When it comes to practice, there is a gap in our capabilities. For example, even simple manned-missions to Mars would require sending much heavier payloads than we have to Mars so far. These spacecrafts would have to travel

at high speeds towards Mars, then decelerate rapidly enough to safely land the people on the surface. So far, not a single mission has demonstrated the capability of executing a soft landing with heavy payloads. Terraforming the planet involves solving a series of such problems which we have not tackled before.

INTRODUCING AN ATMOSPHERE

Carl Sagan was the first to suggest a practical way in which an atmosphere could be introduced to Mars, by vaporising the carbon dioxide ice and water ice trapped beneath the northern polar ice cap. The exact process suggested included using genetically-modified plants that could survive in the extreme Martian conditions to darken the surface and melt the polar ice caps. A series of channels dug towards the equator would allow the molten water to flow to the lower latitudes. The end goal was to increase both the atmospheric pressure and temperature on Mars. This was a suggestion first proposed in 1971.

Mars lost its magnetosphere more than three billion years ago. This magnetosphere protected the atmosphere